

Lab 3: SQL

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1 Objectives

The purpose of this lab is to create a full working database using SQL. You must use the Data Definition Language (DDL) to create tables and the Data Manipulation Language (DML) to insert data and extract information from your tables.

2 Reading

To be able to actively participate in this lab the student is expected to have read the lectures (obviously) and participated to the workshops (not mandatory but a very good idea). Additional reading from provided resources is beneficial but not crucial to the completion of this lab.

3 Setup

For this lab, you can use your connection to JTH MSSQL Server (from the lab computers at JTH, see lecture 03) **or** MariaDB (running with Docker on your local computer, see lecture 00).

4 Presentation

The presentation for Lab 3 is done orally. **All group members** present to the lab assistant which then determines if the demonstrated knowledge is sufficient for a pass. If not, the students can present at a later occasion, same or other day. The lab is graded individually, meaning that one team member can pass while the other fails.

5 Instructions

You must work in a **group of 2 students**, together, but with equal work distribution. Your team will be the same for all the remaining laboratory assignments as well. You should be with a partner that is in the same lab session as you (easier for everybody).

Remarks:

- With your JTH MSSQL Server account, you **cannot create a database**; you only have the right to create tables in your SQL Server environment. It is not a problem, please just create tables in your database.
- For this lab, you **must** create a text file containing your SQL instructions to create, insert data and extract information from your tables (see point 6 in the scenario).

6 Scenario

In this exercise, you are **FREE** to choose any subject you want: a garage, a shop selling things; a music, a bank, a movies, a super hero database... Please do the described work as a group of two students:

1) Imagine and create an Entity-Relationship Diagram with **five entities** and at least **four relationships**, add the cardinalities to your Diagram. You must create a **3NF database** to avoid data redundancy and anomalies (insertion, deletion, update anomalies). You can use any tool you want to create your ER Diagram (draw.io, pen and paper, etc.). You must create a PDF file with your ER Diagram and submit it on Canvas.

2) Transform this ER Diagram into a set of tables following the rules given in lecture 01. Make the tables in **3NF** by following the rules given in lecture 03. You must create a PDF file with your ER Diagram and tables and submit it on Canvas.

3) Create ALL the tables on a DBMS (MSSQL Server if it works **or** MariaDB with Docker). Each table **MUST** have at least 10 rows! Put ALL the SQL code in a text file and submit this text file for lab 3 on Canvas.

4) Create 10 (working) queries on your data and EXPLAIN why they are useful for your database:

- a. 6 SELECT ... FROM ... with WHERE conditions
- b. 2 DELETE FROM ... WHERE conditions
- c. 2 UPDATE ... WHERE conditions

These queries **MUST BE** present in the SQL text file with your SQL CODE (see point 3 above).

5) Present your database to and answer the questions from the lab Assistant.

6) Create a text file with extension ".sql" with ALL the necessary SQL instructions to create (CREATE TABLE ...) your tables, insert data (INSERT INTO ...) into them and extract/modify/remove (SELECT ..., UPDATE ..., DELETE ...) the information you want from them. Put this text file on Canvas in the lab3 assignment.

You need to copy and paste **ALL your SQL instructions** in a text file WHILE (on the fly) creating your tables and data. Once you close your connection to JTH MSSQL Server or MariaDB with Docker, you won't be able to find the code again!